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EXPERIMENT 3

Aim: To implement lemmatization using spacy and NLTK.

Theory:

Lemmatization: Lemmatization is the process of reducing a word to its base or root form, known as the lemma. Unlike stemming, which simply cuts off prefixes or suffixes, lemmatization considers the context and applies proper linguistic rules to produce a valid word.

Lemmatization in NLP: In Natural Language Processing (NLP), lemmatization is a crucial step for text pre-processing. It involves converting words to their base or root form (the lemma) while retaining the correct meaning. This helps to standardize words so that variations (like plural forms, tenses, or inflections) can be treated as a single entity during analysis.

Code:

1)

```
import nltk
nltk.download('wordnet')

from nltk.stem import WordNetLemmatizer

lemmatizer = WordNetLemmatizer()

print("rocks :", lemmatizer.lemmatize("rocks"))
print("corpora :", lemmatizer.lemmatize("corpora"))
print("worse :", lemmatizer.lemmatize("worse", pos='a'))

print("better :", lemmatizer.lemmatize("better", pos="a"))
```

2)

```
import spacy

nlp = spacy.load('en_core_web_sm')

text='The quick fox are jumping over the lazy dogs.'

doc = nlp(text)
```



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```
lemmatized_tokens = [token.lemma_ for token in doc]

lemmatized_text = ' '.join(lemmatized_tokens)

print("Original text: ", text)
print("Lemmatized text: ", lemmatized_text)
```

Output:

1)

```
rocks : rock
corpora : corpus
worse : bad
better : good
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
```

2)

```
➡ Original text: The quick fox are jumping over the lazy dogs.
Lemmatized text: the quick fox be jump over the lazy dog .
```

Conclusion: Thus we successfully implemented lemmatization using spacy and NLTK.